Introduction to Artificial Intelligent (AI)

1. Course Introduction



Course Objectives

This course is designed to provide student with:

- Conceptual understanding of the technology behind Artificial Intelligent
- Ability to implement various AI system
- Ability to apply AI to solve complex real-world problems

Course Outline

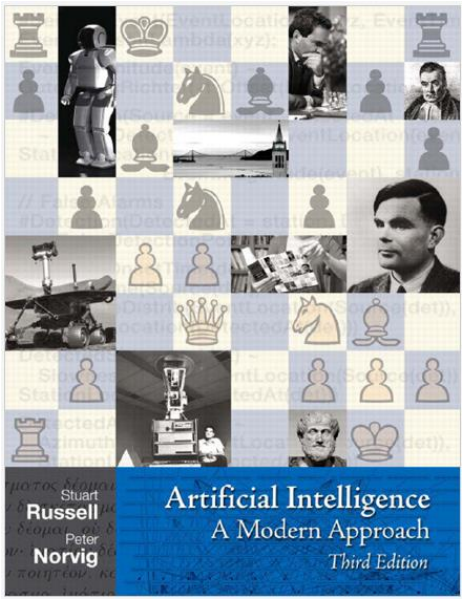
- Introduction to AI
  - What is AI
  - Main areas of AI
  - History of AI
- Algorithms
  - Searching algorithms
  - Motion/Action planning algorithms
- Probabilistic reasoning in AI
  - Uncertainty and decision
  - Bayesian networks

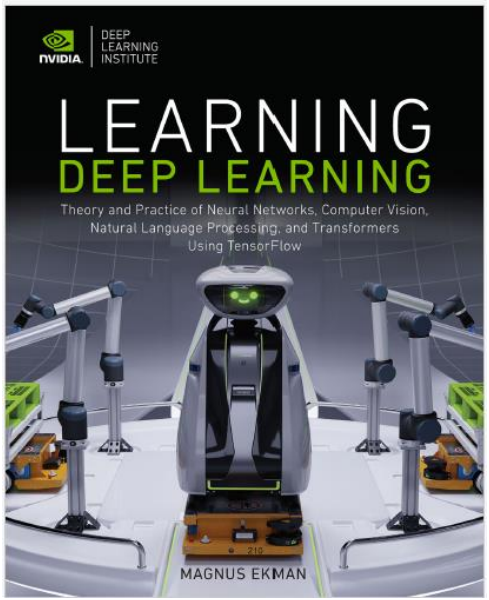
Course Outline

- Machine learning
  - Supervised learning
  - Unsupervised learning
- Deep Learning
  - Convolution neural network
  - Recurrent network
  - Attention and transformer

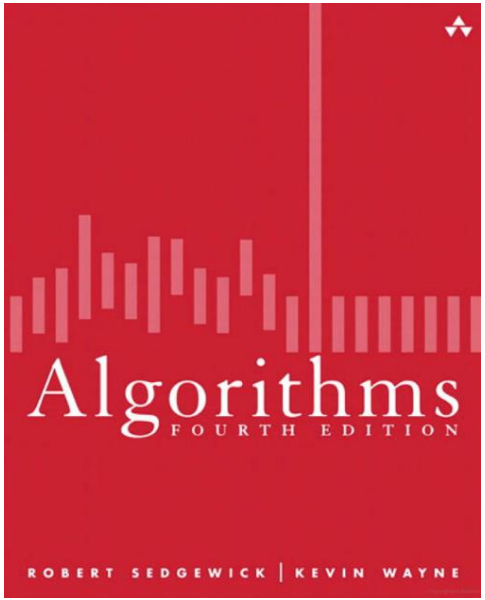
References

Textbooks





Reference



Software

1. **Miniconda:**

   https://docs.conda.io/en/latest/miniconda.html

   Create new environment and install python, numpy, matplotlib, scikit learn



2. **VSCode:**

   https://code.visualstudio.com/



3. **Google Colab:**

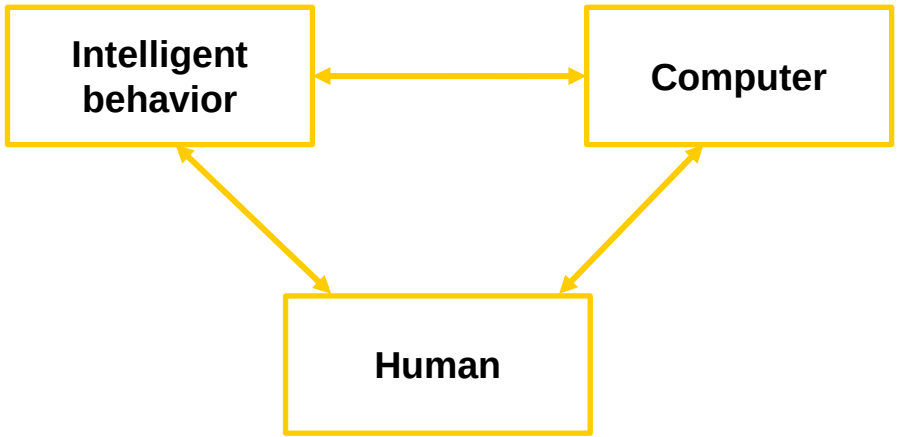
   https://colab.google/



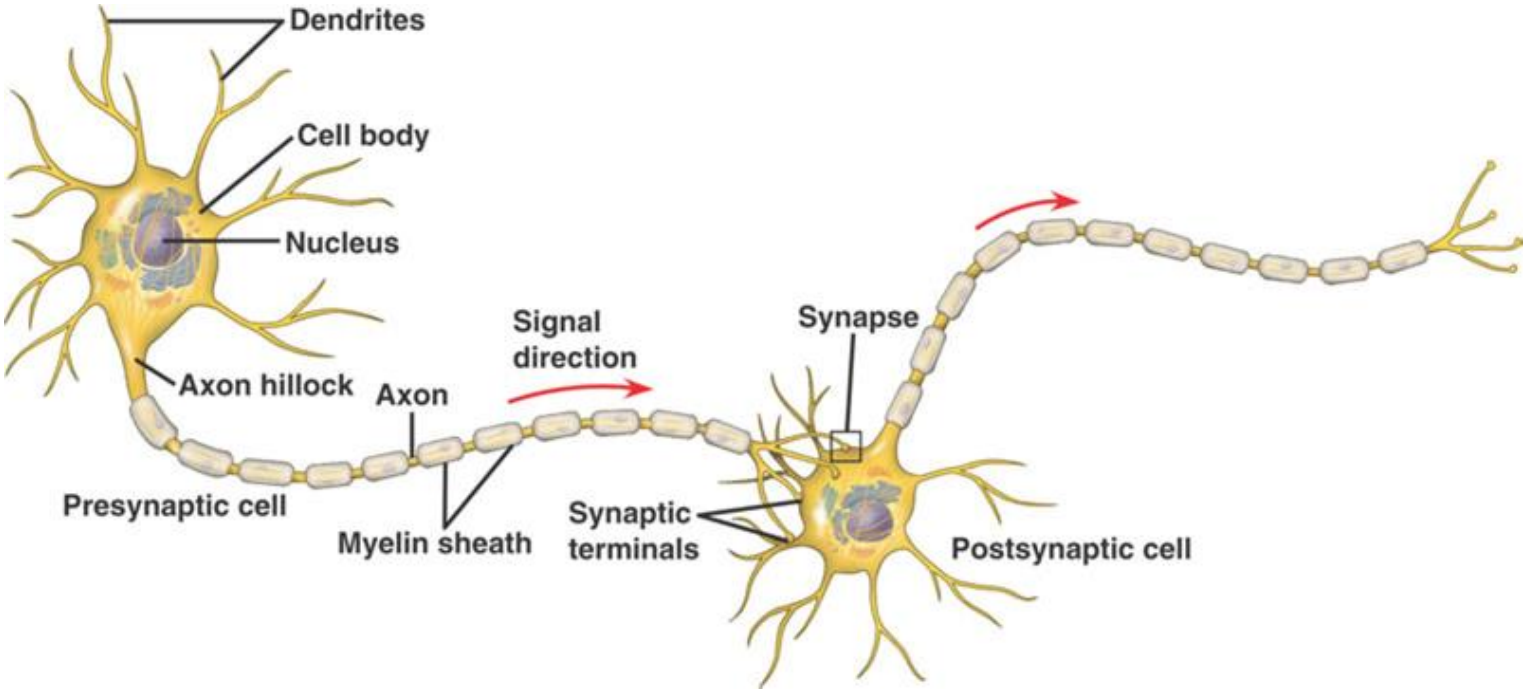
What is AI ?

an attempt of

AI is ~~the~~ reproduction of **human reasoning and intelligent behavior** by **computational methods**



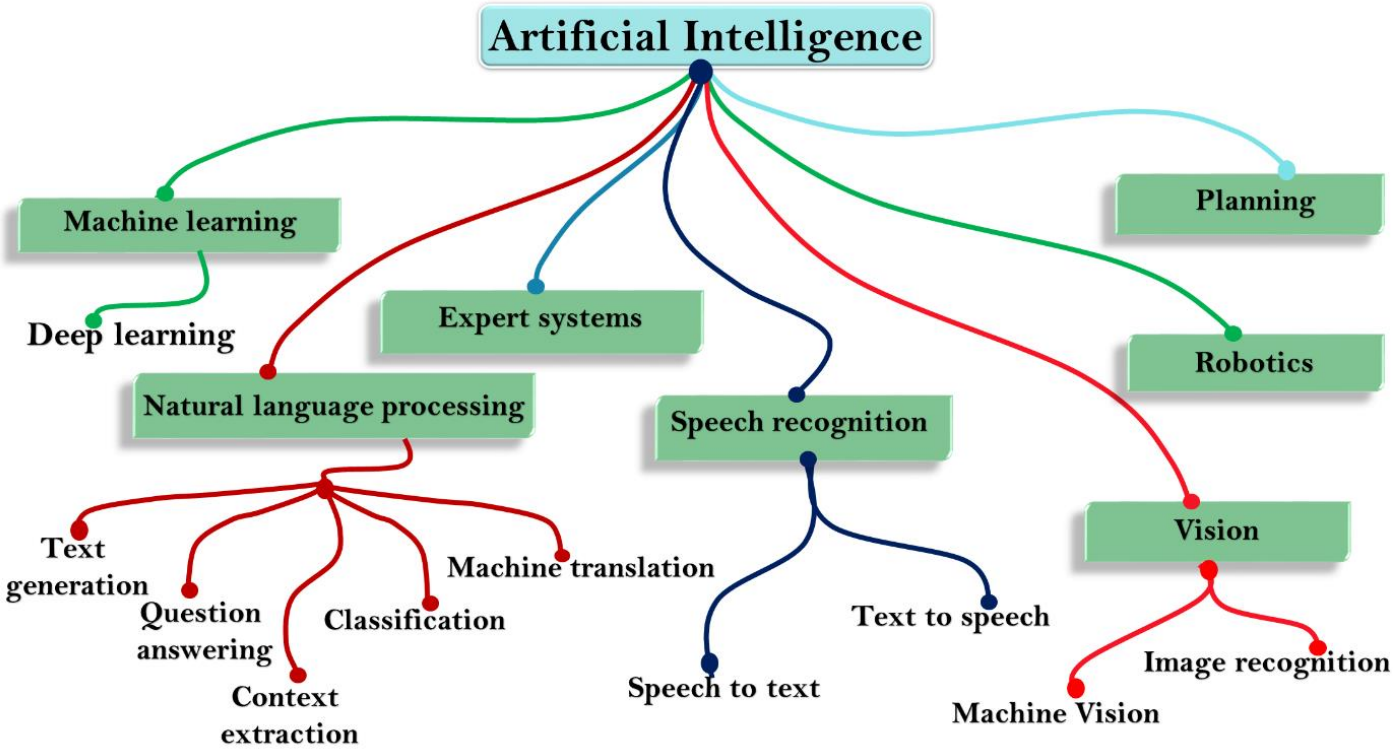
Biological Neuron



History of AI

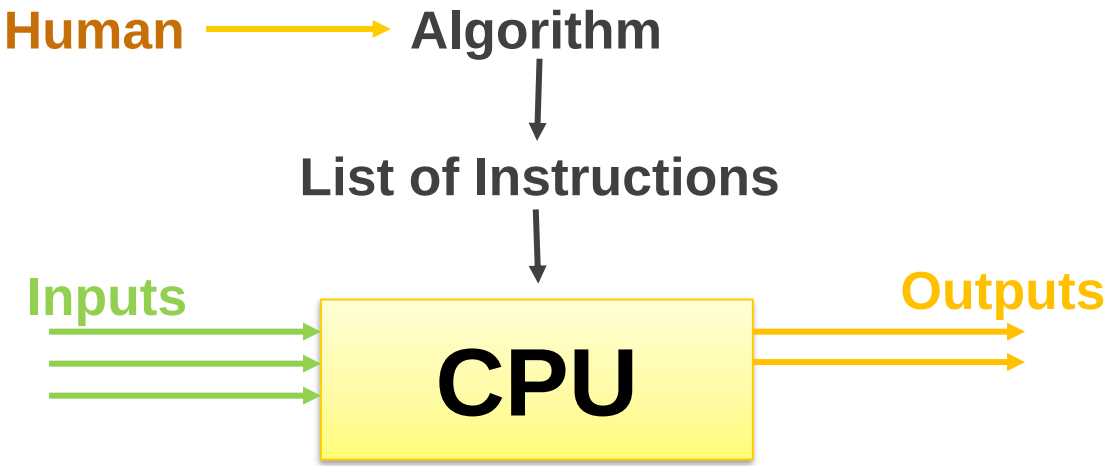
1956: The name “Artificial Intelligence” is coined
1960’s: Search and games, formal logic and theorem proving
1970’s: Robotics, perception, knowledge representation, expert systems
1980’s: More expert systems, AI becomes an industry
1990’s: Rational agents, probabilistic reasoning, achine learning
2000’s: Systems integrating many AI methods, machine learning, reasoning under uncertainty, robotics again
2010’s: Deep learning and big data
2020’s: Attention and Transformer, Generative AI

Main Areas of AI



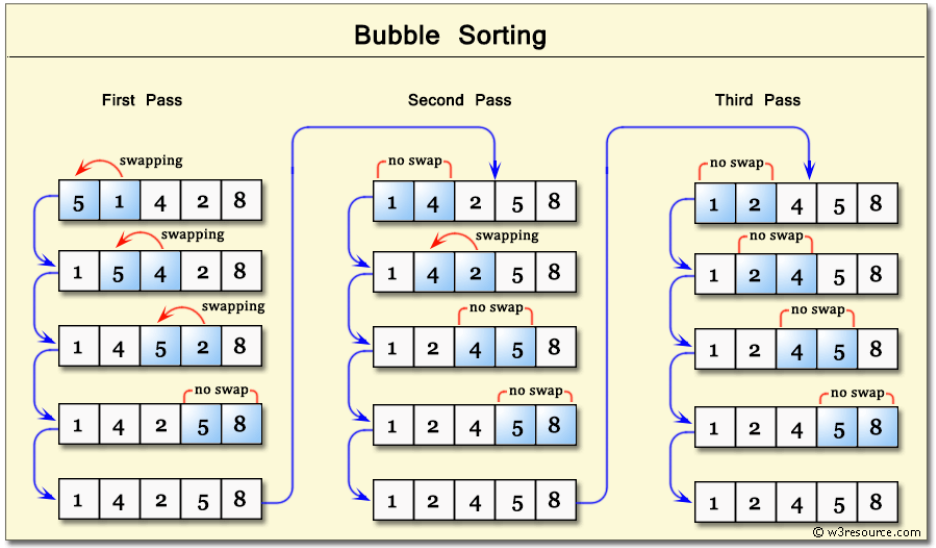
Algorithm

An **algorithm** is **a procedure** used for solving a problem or performing a computation



Algorithm

Example: sort numbers (5,1,4,2,8) in ascending order



Algorithm 1: Bubble sort

```
Data: Input array A[]
Result: Sorted A[]
int i, j, k;
N = length(A);
for j = 1 to N do
    for i = 0 to N-1 do
        if A[i] > A[i+1] then
            temp = A[i];
            A[i] = A[i+1];
            A[i+1] = temp;
        end
    end
end
```

AI agent

AI Agents

An **AI agent** is a computer program that is designed to perceive its environment, make decisions and take actions to achieve a specific goal

An **AI agent** operates autonomously

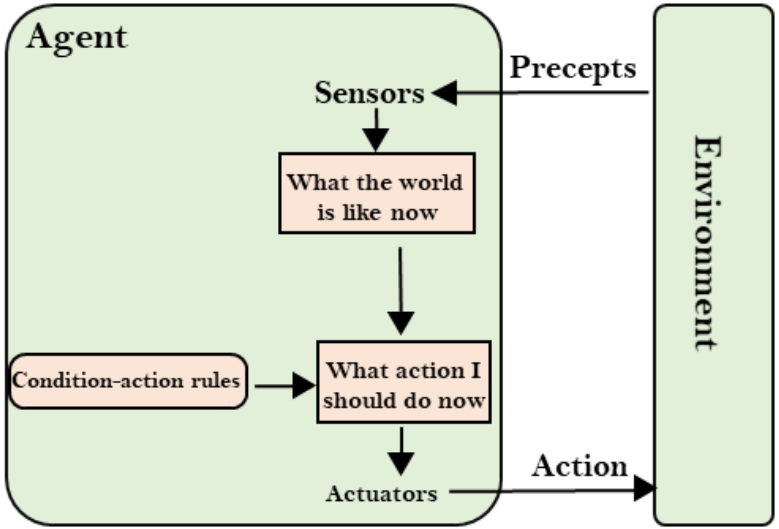
Types of AI agent:

- Simple Reflex Agent
- Model-based reflex agent
- Goal-based agents
- Utility-based agent
- Learning agent

AI Agents

Simple Reflex Agent

- The **Simple reflex agents** act only on the basis of the current percepts and ignore the rest of the percept history.
- These agents only work in the fully observable environment.
- The Simple reflex agent works on Condition-action rule.
- Ex: Mail sorting Robot

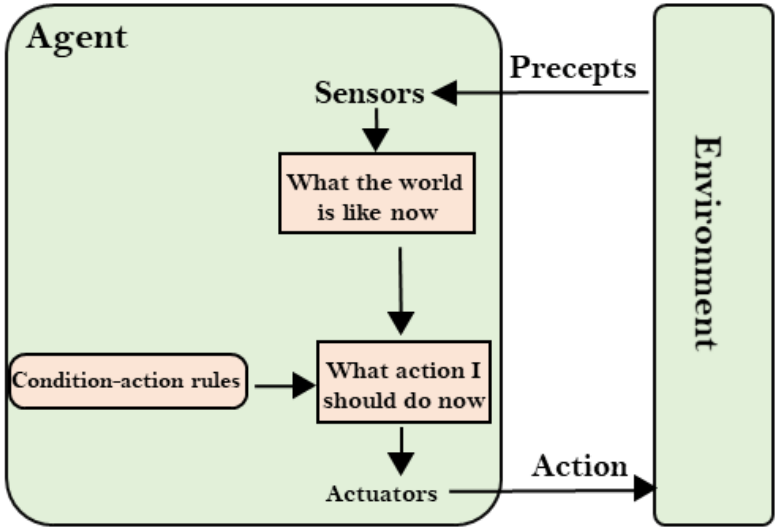


AI Agents

Simple Reflex Agent

Mail sorting robot

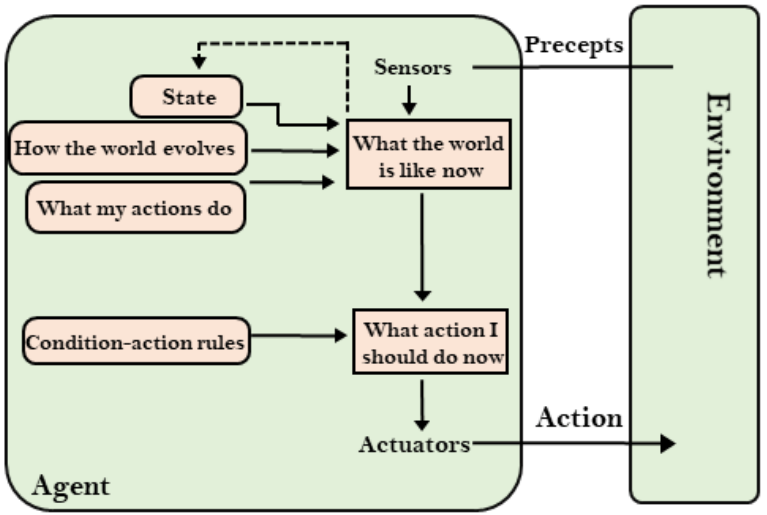
- Agent: sorting robot
- Environment: conveyor belt of letter
- Rule: e.g. address=Q1 → put into Q1 bag



AI Agents

Model-based Reflex Agent

- The **Model-based reflex agents** use internal memory and a percept history to create a model of the environment.
- It works by finding a rule whose condition matches the current situation
- These agents can work in the partially observable environment.
- Ex: Vacuum cleaner robot

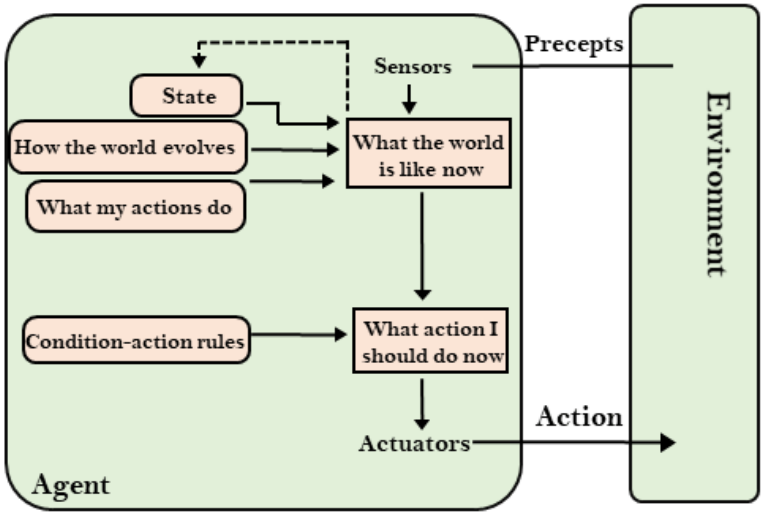


AI Agents

Model-based Reflex Agent

Vacuum cleaner robot

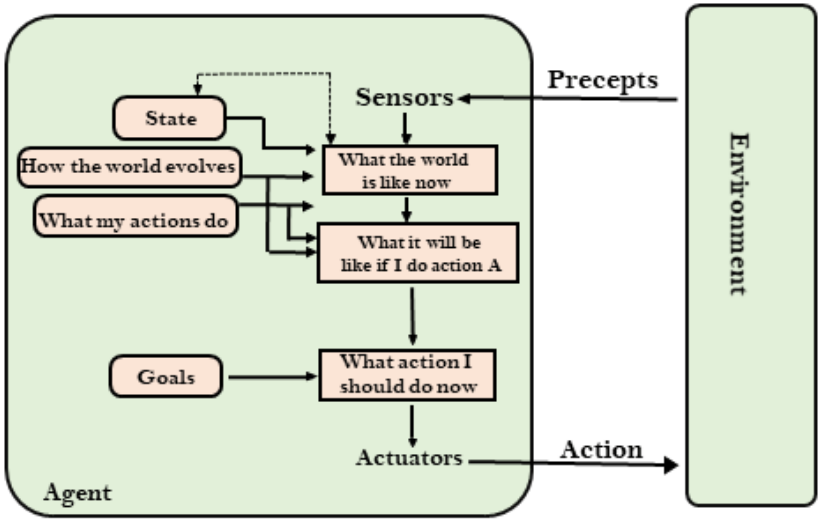
- Agent: cleaner robot
- Environment: room, furniture
- Model: map of the room, areas already cleaned



AI Agents

Goal-based agent

- Knowledge of the present state isn't always sufficient to make a decision → Goals
- Goal-Based Agents choose actions to achieve the goal by searching and planning
- Ex: Self Driving Car

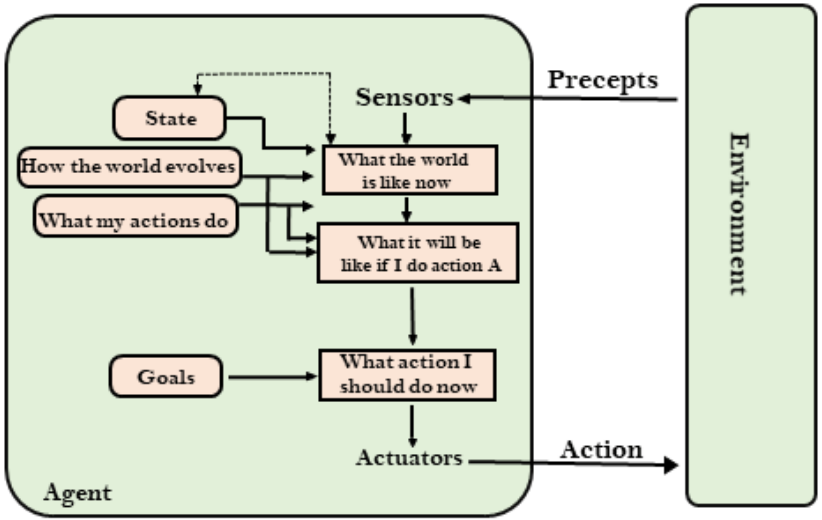


AI Agents

Goal-based agent

Self Driving Car

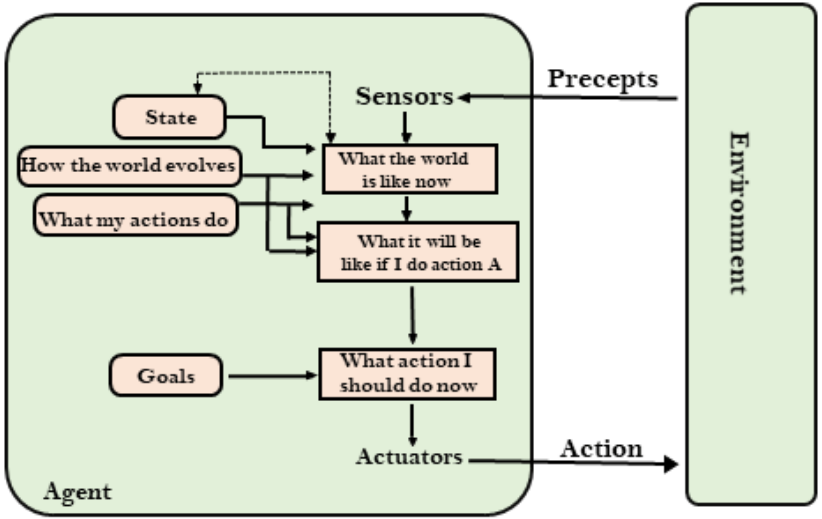
- Agent: autonomous car
- Environment: street, vehicles, pedestrians, traffic signal/sign
- Goals: safe, fast, legal trip
- Percept: camera, GPS, speedometer, sonar
- Actions: steer, accelerate, brake



AI Agents

Utility-based agent

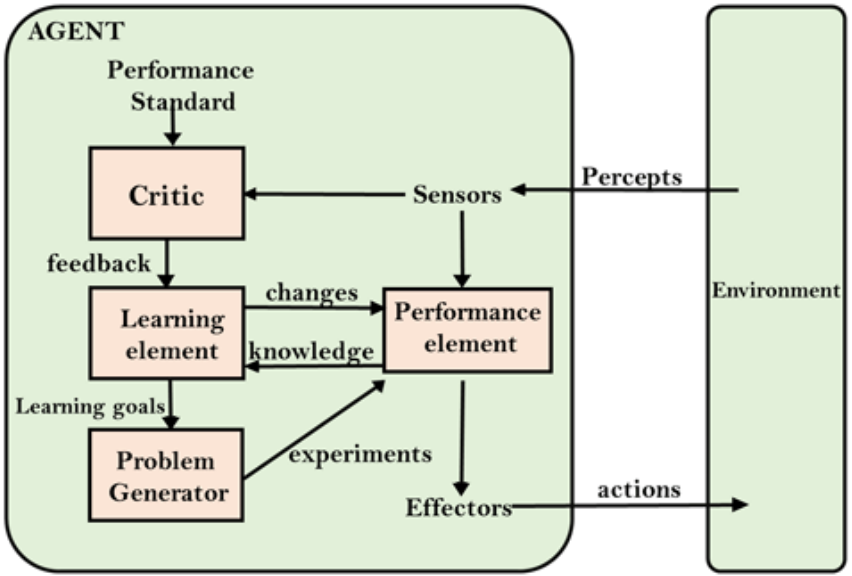
- **Utility:** measure of goodness
- **Utility-based agent** = Goal-based agent + optimise utility over range of goals
- Ex: Self Driving Car



AI Agents

Learning agent

- A **learning agent** is the type of agent which can learn from its past experiences.



AI Agents

What sort of AI agent would a chess playing program is?



Deep Blue defeat chess champion Garry Kasparov - 1997